

corrigé zakito

Énoncé

1

a

$$g(t) = \ln(1-t) + t + \frac{t^2}{2(1-t)}, \quad g(0) = 0.$$

$$g'(t) = -\frac{1}{1-t} + 1 + \frac{2t-t^2}{2(1-t)^2} = \frac{t^2}{2(1-t)^2} \geq 0 \quad (0 \leq t < 1).$$

$$\ln(1-t) + t + \frac{t^2}{2(1-t)} \geq 0.$$

b

$$t = \frac{1}{p} \Rightarrow \frac{1}{p} \geq -\ln\left(1 - \frac{1}{p}\right) - \frac{1}{2p(p-1)}.$$

$$\sum_{p \leq n} \frac{1}{p} \geq \ln \prod_{p \leq n} \left(1 - \frac{1}{p}\right)^{-1} - \frac{1}{2} \sum_{p \leq n} \frac{1}{p(p-1)}.$$

2

$$\sum_{k=0}^r \frac{1}{p^k} = \frac{1-p^{-r-1}}{1-p^{-1}} \leq \left(1 - \frac{1}{p}\right)^{-1}.$$

$$R = \left\lfloor \frac{\ln n}{\ln 2} \right\rfloor, \quad 1 \leq m \leq n \Rightarrow m = \prod_{p \leq n} p^{\alpha_p(m)}, \quad 0 \leq \alpha_p(m) \leq R.$$

$$\prod_{p \leq n} \left(1 - \frac{1}{p}\right)^{-1} \geq \prod_{p \leq n} \sum_{k=0}^R p^{-k} = \sum_{(\alpha_p) \in \{0, \dots, R\}^{\{p \leq n\}}} \frac{1}{\prod_{p \leq n} p^{\alpha_p}} \geq \sum_{m=1}^n \frac{1}{m}.$$

3

a

$$t \in [k, k+1] \Rightarrow \frac{1}{t} \leq \frac{1}{k} \Rightarrow \int_k^{k+1} \frac{dt}{t} \leq \frac{1}{k}.$$

$$\sum_{k=1}^n \frac{1}{k} \geq \int_1^{n+1} \frac{dt}{t} = \ln(n+1) \geq \ln n.$$

b

$$\sum_{p \leq n} \frac{1}{p(p-1)} \leq \sum_{k=2}^n \left(\frac{1}{k-1} - \frac{1}{k} \right) = 1 - \frac{1}{n} \leq 1.$$

$$\sum_{p \leq n} \frac{1}{p} \geq \ln \left(\sum_{k=1}^n \frac{1}{k} \right) - \frac{1}{2} \geq \ln \ln n - \frac{1}{2}.$$

4

a

$$Q_n = \prod_{n+1 < p \leq 2n+1} p, \quad n! \binom{2n+1}{n+1} = (n+2)(n+3) \cdots (2n+1).$$

$$\forall p \mid Q_n, \quad p \mid n! \binom{2n+1}{n+1}, \quad p \nmid n! \Rightarrow p \mid \binom{2n+1}{n+1}.$$

$$Q_n \mid \binom{2n+1}{n+1}, \quad 2 \binom{2n+1}{n+1} = \binom{2n+1}{n} + \binom{2n+1}{n+1} \leq \sum_{k=0}^{2n+1} \binom{2n+1}{k} = 2^{2n+1}.$$

$$Q_n \leq 4^n.$$

b

$$P_N = \prod_{p \leq N} p, \quad P_0 = P_1 = 1, \quad P_2 = 2.$$

Récurrance forte :

$$N = 2m + 1 \geq 3 \Rightarrow P_N = P_{m+1} Q_m \leq 4^{m+1} 4^m = 4^N.$$

$$N = 2m \geq 4 \Rightarrow N \notin \mathbb{P} \Rightarrow P_N = P_{N-1} \leq 4^{N-1} \leq 4^N.$$

$$\forall N \in \mathbb{N}, \quad P_N \leq 4^N.$$

5

a

$$2 \leq k \leq n, \quad t \in [k-1, k] \Rightarrow \ln t \leq \ln k.$$

$$\ln(n!) = \sum_{k=2}^n \ln k \geq \int_1^n \ln t \, dt = n \ln n - n + 1 \geq n \ln n - n.$$

b

$$p_k \geq k \Rightarrow n! \leq \prod_{k=1}^n p_k = P_{p_n} \leq 4^{p_n}.$$

$$n \ln n - n \leq \ln(n!) \leq 2p_n \ln 2 \Rightarrow p_n \geq \frac{n \ln n - n}{2 \ln 2}.$$

6

$$\zeta_{n+1}(s) - \zeta_n(s) = (n+1)^{-s} > 0.$$

$$\sum_{k=2}^n k^{-s} \leq \int_1^n t^{-s} \, dt = \frac{1 - n^{1-s}}{s-1} \leq \frac{1}{s-1}.$$

$$\zeta_n(s) \uparrow \zeta(s) \leq 1 + \frac{1}{s-1}.$$

7

a

$$r = |\{p \in \mathbb{P} : p \leq n\}|, \quad T = \left\lfloor \frac{\ln n}{\ln 2} \right\rfloor.$$

$$1 \leq m \leq n \Rightarrow m = \prod_{i=1}^r p_i^{\alpha_i}, \quad 2^{\alpha_i} \leq p_i^{\alpha_i} \leq m \leq n \Rightarrow 0 \leq \alpha_i \leq T.$$

$$\{1, \dots, n\} \subset E_{r,T}.$$

$$p_r \leq n, \quad p_r \notin E_{r-1,t} \quad (\forall t) \Rightarrow r \text{ minimal.}$$

b

$$\prod_{i=1}^r \sum_{j=0}^t p_i^{-sj} = \sum_{\alpha_1, \dots, \alpha_r=0}^t (p_1^{\alpha_1} \cdots p_r^{\alpha_r})^{-s} = \sum_{m \in E_{r,t}} m^{-s}.$$

$$t \geq T \Rightarrow \{1, \dots, n\} \subset E_{r,t} \subset \mathbb{N}^* \Rightarrow \zeta_n(s) \leq \prod_{i=1}^r \sum_{j=0}^t p_i^{-sj} \leq \zeta(s).$$

c

$$t \rightarrow +\infty : \quad \zeta_n(s) \leq \prod_{i=1}^r \frac{1}{1 - p_i^{-s}} = \pi_n(s) \leq \zeta(s).$$

$$\zeta_n(s) \rightarrow \zeta(s) \Rightarrow \pi_n(s) \rightarrow \zeta(s).$$

8

$$h(t) = t - \ln(1+t), \quad h'(t) = \frac{t}{1+t}, \quad h(0) = 0.$$

$$h \text{ décroissante sur } (-1, 0], \text{ croissante sur } [0, +\infty) \Rightarrow h(t) \geq 0.$$

$$\ln(1 - p^{-s}) \leq -p^{-s} \Rightarrow \sum_{p \leq n} p^{-s} \leq -\sum_{p \leq n} \ln(1 - p^{-s}) = \ln \pi_n(s) \leq \ln \zeta(s).$$

9

a

$$s = 1 + \frac{a_n}{\ln n} > 1, \quad p \leq n \Rightarrow p^{s-1} \leq e^{a_n}.$$

$$\sum_{p \leq n} \frac{1}{p} = \sum_{p \leq n} \frac{p^{s-1}}{p^s} \leq e^{a_n} \sum_{p \leq n} p^{-s} \leq e^{a_n} \ln \zeta(s) \leq e^{a_n} \ln \left(1 + \frac{\ln n}{a_n} \right).$$

b

$$n \geq 3, \quad L_n = \ln \ln n > 0, \quad a_n = \frac{1}{L_n} \rightarrow 0.$$

$$\ln(1 + L_n \ln n) = L_n + \ln L_n + \ln \left(1 + \frac{1}{L_n \ln n} \right) = L_n + o(L_n).$$

$$1 - \frac{1}{2L_n} \leq \frac{1}{L_n} \sum_{p \leq n} \frac{1}{p} \leq e^{1/L_n} \frac{\ln(1 + L_n \ln n)}{L_n} \rightarrow 1.$$

$$\sum_{p \leq n} \frac{1}{p} \sim \ln \ln n.$$